Session 9. Vertical Integration, Outsourcing, Joint Ventures Wahaha-Danone

Competitive Strategy

Luis Garicano

Vertical Integration: The Question

- Make or Buy?
  - Should Booth operate cafeteria or outsource?
  - Should Ford own Hertz?
- Forward integration: a firm performs activities typically done by its customers (downstream firms)
- Backward integration: a firm performs activities typically done by its suppliers (upstream firms)

Make-or-Buy Fallacies

- Fallacy 1: Vertical integration is better because we can obtain the activity "at cost", avoid supplier mark-up.
  - In perfectly competitive market, price equals long run average cost
    - Can only do better if own costs lower than markets
  - If market is not competitive, must understand why
    - Why can you (not others) overcome barriers to entry
    - Why is market not working efficiently?
  - => Would your competitor want to buy from you?

Make-or-Buy Fallacies

- Fallacy 2: Integration to obtain the other firm's profit.

- High attractiveness as a reason to enter is often problematic argument
  - Why is it attractive?
  - Why don't more competitors enter then
  - Will it stay attractive

- BUT… not impossible if there are barriers to entry that only few can scale

Rationales for (or against) vertical integration

- Economics of scale and scope (mostly against V.I.)
- Incentives: Trade-off between
  - Strong market incentives (but only for what market rewards!)
  - Weaker, but often more balanced incentives under integration

- Hold-up and relation-specific investments (mostly in favor of V.I.)
- Legal considerations & others
  - Tax avoidance: Using transfer prices to shift revenues (in favor)
  - Avoiding anti-trust regulations on RPM, exclusive distribition (in favor)

Economics of Scale and Scope

- Economies of Scale and Scope:
  - Supplier/buyer firms can achieve more economies of scale in production:
    - In-house department produces less than market specialist
      - Booth outsources cafeteria, cleaning services,…
      - Disney might outsource running hotels.
  - Do you have a competitive advantage in this part of the value chain? Would your competitor want to buy from you?

Incentives (I)

- Divisions within firms do not face same incentives as market firms
  - Suppliers have strong incentives to improve efficiency
    - Can switch among suppliers; market will tend to drive out those that are inefficient or don't innovate
  - Harder to motivate managers within divisions
    - Absence of market discipline: often guaranteed in-house orders
    - Head-quarters always loom large in key decision
      - Weaker accountability
      - Lengthy authorization stifles initiative and motivation

INCENTIVES (II)

Trade-off:

- Market: strong, but possibly unbalanced incentives.
  - Partner has incentives to do what the market rewards → also market suffers from YGWYPF

- Internal production: Weak, but more balanced incentives.
  - Benefits of low powered incentives
  - Weakening market incentives may lead to a better allocation of effort toward things markets do not directly reward

INCENTIVES (III)

- What you should outsource depends on what markets can or cannot easily reward directly
- How easy is it to write a formal contract specifying performance?
- Difficult things tend to be user-specific, intangible, service-oriented, difficult to specify in advance, etc.
- Difficult things often involve "coordination", "cooperation"
  - Example: Bombardier and Morrison Knudsen in railcars

INCENTIVES (IV)

- Outsource only when “performance” can be easily specified and monitored

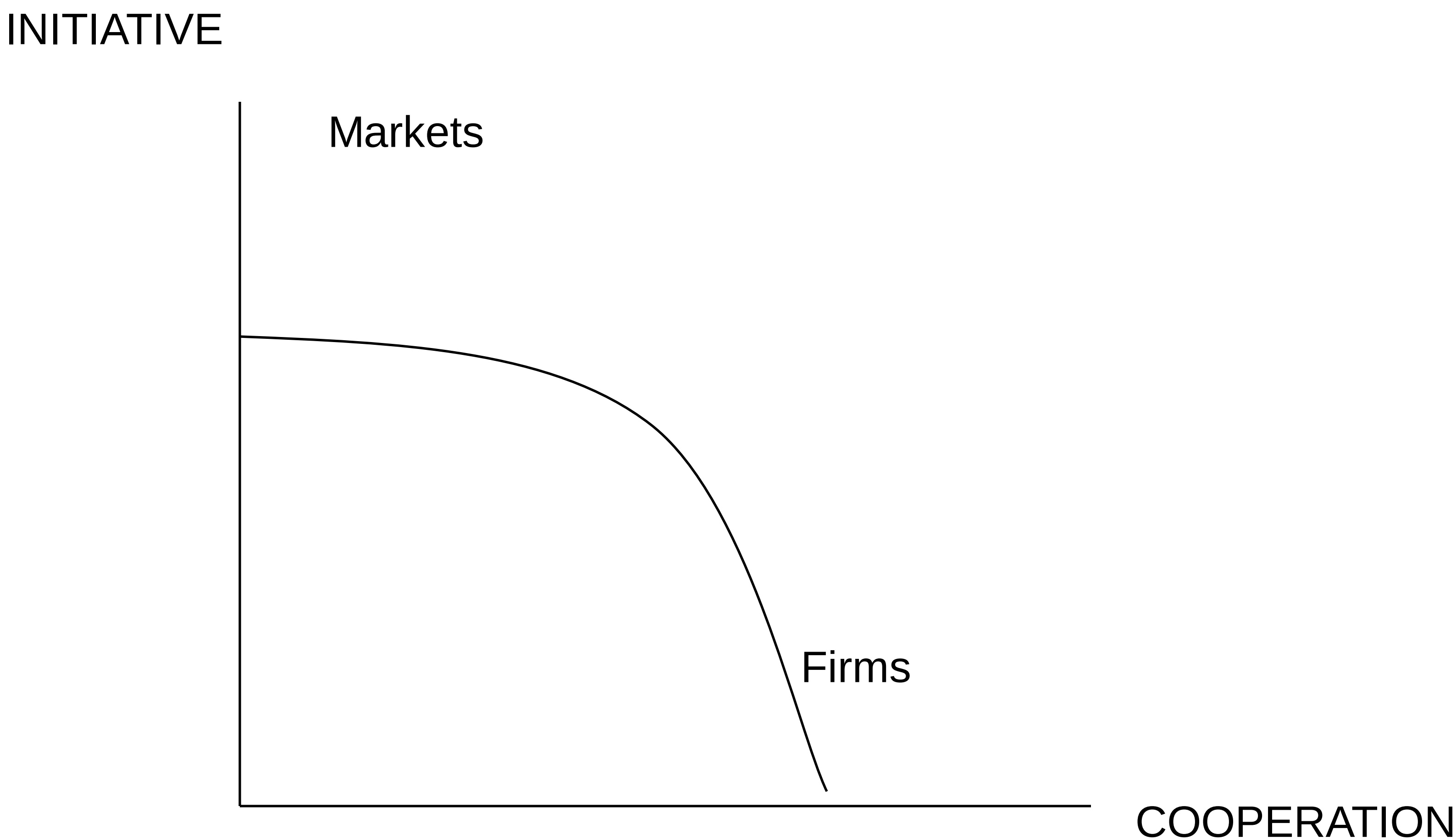
- If activity hard to measure, performance rewards tricky
  - In the latter case, use the firm, use direct monitoring/hierarchy together with subjective rewards, long term and implicit contracts, etc.
    - thus firm associated with limited incentives, wage contracts etc.

INITIATIVE VS. COORDINATION

- Use market: provide strong rewards on the margin to encourage initiative taking
  - but risk lack of cooperation

- or use ‘firm’ /bureaucracy–not so strong rewards
  - obtain coordination but at the risk of not rewarding effort

INITIATIVE VS COORDINATION



Incentives: Examples

- Sales : Three activities potentially important in sales
  - Making current sales (1)
  - Cultivating long-term customer satisfaction (2)
  - Gathering information about customer needs (3)

  - If only selling activities are important: independent sales agent
    - high power incentives – pay a commission per sale
    - allow non exclusivity (no problem, motivated by commission)

  - If non-selling activities (2), (3) important – employee agents
    - pay all in wage, low-powered incentives
    - not allow to sell other goods

INCENTIVES: EXAMPLES (III)

- Need for rapid / flexible coordination
  - If input needs are complex & changing, constant negotiation / haggling can be very costly
  - Difficult to provide incentives up front for suppliers to respond in an efficient way
    - Example: ZARA's quick response business model

- Can contracts solve coordination problems? Often they can:

→ Airline Alliances

RECALL: HOLD-UP AND RELATION-SPECIFIC INVESTMENTS

- Hold-up is an important reason for vertical integration
- Key issue in Business-to-Business, but also important within organizations.
  - When two companies start working with each other, there is often a “fundamental transformation” from a competitive relationship to a specific relationship. The parties become dependent on each other.
- Basic idea of hold-up
  - I ask to renegotiate an earlier (implicit or explicit) agreement in order to increase my share of the pie.
  - If the contract is not water-tight, you often have not much choice but accepting new terms.

RECALL: HOLD-UP

- Conditions that make hold-up likely/important
  - Asset Specificity – Relationship-specific investments
    - A specialized asset is tailor-made for one or a few specific buyers
      - Opportunities for exploitation are greater when one firm is wholly dependent on another
    - Specific physical capital, specific human capital, site-specific capital
  - Uncertainty and contracting difficulties
  - Infrequent relation

Hold-up (II)

- Classic Hold-up Example: GM and Fischer-Body

- In stable environment, “easy” to write contract about quantity, price and quality of car bodies Fischer should supply to GM

  - E.g. 2000 bodies a day, of a specified type, for the foreseeable future

- But what if changes in the environment?

  - Shock to demand for GM, shock to cost to Fischer-Body, new regulations on car pollution, new trade

Hold-up: Contracts vs Integration

In theory, hold-up could avoided through *optimal* arm length's contracts

- Optimal contracts do not need to be renegotiated
- However, writing such optimal contracts is costly
  - Difficult to think ahead, describe all possible contingencies and actions to be taken in each case

Inability to specify all outcomes, verify actions create potential for *opportunistic behavior* when contracts have to be revised

- Each side may try to interpret the terms of a contract to its advantage, especially when terms are vague or even missing

Hold-up: Contracts vs Integration

The hold-up problem raises transaction costs

- More difficult contract negotiations and renegotiations, distrust:
  - costs associated with trying to write better contracts, costs of disputing contracts, renegotiating.

- Reduced investment in relationship-specific investments
  - Refusal to make “tailored ” or “customized” investments for a buyer

Vertical integration reduces these transaction costs

Integration gives one agent control over assets on both sides and eliminates the hold-up problem

Hold-up: Examples

Other Examples

- Make-or-buy in the aerospace industry, car industry
  - High design specificity inputs more likely to be produced in-house
- Backward integration in can making by Beer and Food Producers
- Coal Mines and power plants

EMPIRICAL EVIDENCE

- Strong evidence that vertical integration is more likely when potential hold-up problems are big
  - e.g. Quasi-integration by automobile manufacturers more likely when specialized machines cannot be used to produce for other manufacturers
  - “Design-specificity” in airline components predicts vertical integration
- Evidence that inputs which are “complex” or require intensive coordination more likely to be produced in-house

MAKE OR COOPERATE

- Long-run relationship with exclusive (or small number of) suppliers may be a substitute for In-House production

  - Exclusivity provides incentives to supplier to make relationship-specific investments.
    - Less risk of hold-up by buyer which is now more dependent on supplier.
    - Increases the frequency of the interaction

  - Repeated interaction mitigates risks of hold-up by supplier or buyer.

Make or cooperate: Alternatives to Vertical Integration

- A variety of in between situations can try to capture the best of both worlds
  - Tapered integration
  - Joint ventures and strategic alliances
  - Subcontractors networks & Japanese Keiretsu
- The crucial element in JVs, alliances, Keiretsusis the relational contract aspect
  - We cooperate because we expect the other side to cooperate
  - The existence of a future is thus key
  - And the partners have to shape situation on the ground to attain their objectives
    - Role of the law is limited, particularly in developing world
    - Is all about TRUST, and not about what is written in the contract!

1. Tapered Integration

- Mixture of vertical integration and market exchange = make AND buy
- A firm may produce part of its input on its own and purchase the rest
- A firm may sell part of its output through in-house sales efforts and sell the rest through independent distributors

Examples of Tapered Integration

- Hotel, video, hamburger chains
  - own some of their retail outlets
  - award franchises for others
- Coca-Cola and Pepsi
  - have their own bottling subsidiaries
  - also rely on independently owned bottlers to produce and distribute for some markets
- General Motors
  - has its own market research division
  - also purchases market research from independent firms

Tapered Integration: Advantages

- Additional input/output channels without massive capital investments (helpful for growing firms)
- Information about costs and profitability from internal operations can help in negotiating with market firms
  - threat of self manufacture can discipline external channels, threat to use the market further motivate the performance of its internal channels
- Internal supply capabilities will protect against potential holdups

Tapered Integration: Disadvantages

- Possible loss of economies of scale
- Coordination may become more difficult since two production units must agree on product specifications and delivery times
- A firm's monitoring problems may be exacerbated
  - duplicate efforts of contracting & monitoring
  - mistakenly establish the performance of an inefficient internal supplier as the standard to be met by external suppliers
- Managers may maintain inefficient internal capacity rather than close facilities that had formerly been critical to the firm

2. Strategic Alliances and Joint Ventures

- Alliances involve cooperation, coordination and information sharing for a joint project while the participating firms continue to be independent
- Autonomy
- A joint venture is an alliance where a new independent organization is created and jointly owned by the promoting firms
  - Staffed and operated by employees of one or more parent firms or staffed independently

STRATEGIC ALLIANCE

- Horizontal (same industry), vertical (supplier-buyer) or other
- Alliances and joint ventures are intermediate solutions, between market exchange and vertical integration
- More cooperation, coordination and information sharing than would occur in an arm's length market transaction
  - More later on that
- Rather than rely on contracts, an alliance relies on trust and reciprocity (marriage)
- Disputes must be resolved through negotiation

STRATEGIC ALLIANCE - SCENARIOS

- Uncertainty of future activities prevents the parties from going into the specifics of decisions in contract
- Transactions are complex and one cannot count on contract law to “fill the gaps”
- Existence of relationship-specific assets and potential holdup problem
- Excessively costly for one party to develop the necessary expertise or to carry all of the activities itself
- Market opportunity that induced the transaction is not expected to last very long making a long term contract or merger unattractive
- Regulatory environment necessitates acquiring a local partner for the venture

Strategic Alliance
Toys "R" Us McDonald's Japan

- Start in 1990
- To ease the entry of Toys "R" Us in Japan
- Special law on large retail store: need to be approved before building the stores
- Law protects small merchants
- Toys "R" Us need a local partner
- Choice of McDonald Japan

Strategic Alliance
Toys “R” Us McDonald’s Japan

- Why McDonald’s Japan?
  - President influential politically
  - Good knowledge of Japanese real estate
  - Experience of building McDonald’s Japan
- Deal
  - 20 % stake of McDo Japan in Toys “R” Us Japan
  - 11 stores: 9 would have a McDo inside
- Big advantage: obtain political know-how, site selection expertise and business connections
- Costly or even impossible if alone

Strategic Alliance
Toys "R" Us McDonald's Japan

- Why not sign a contract?
  - Difficulty to measure outcome (efficiency of government lobbying)
  - Poor knowledge of the Japanese situation
  - Could not monitor that McDonald did his job properly
- Fewer incentives problems with the alliances because of the stake in the project
  - IT is about the relational contract!
    - Parties cooperate because they care about future gains
- Less hold up fear for that reason as well

STRATEGIC ALLIANCES: DRAWBACKS

- Leakage of private info
  - Similar as traditional market transactions
  - Risk of losing control over information when collaborating, even worse
- Compromised coordination and time issues
  - No formal mechanism for making decisions or solving disputes
- Free rider problem (monitoring) + influence costs (costs of influence activities for the allocation of resources)

3. Collaborative Relationships

- Japanese industrial firms appear to be less vertically integrated compared to their western counterparts
- Smaller and more specialized in their particular place in the vertical chain
- Japanese firms organize the vertical chain using long term collaborative relationship among firms rather than arm's length transactions
- Two major types of collaborative relationships are found in Japan
  - Subcontractor networks and Keiretsu

SUBCONTRACTOR NETWORKS

- Japanese manufacturers maintain close, informal, long term relationship with their network of subcontractors
- The typical relationship between a manufacturer and a subcontractor involves far more asset specificity in Japan than in the West
  - higher degree of collaboration
  - delegation of more responsibilities to subcontractor

JAPANESE FIRMS

- A typical Japanese firm has about 70 per cent of its shares held by other companies
- The shares are held by a large number of firms in relatively small fractional parcels
- The firm has some kind of transactional relationship with these corporate shareholders (banking, insurance, lending, supply of inputs, purchase of outputs)
- The firm holds shares in many of the these firms: the shareholdings are reciprocal
- Firms hold each other's shares as "stable shareholders" (antei kabunushi)

Keiretsu

- Like a subcontractor network but involves more formalized institutional linkages
- Member firms of a keiretsu hold each other's equity
- Links among firms are further strengthened by personal relationships among top executives
- Most of the key activities in the vertical chain are performed by members of the keiretsu
- 6 largest keiretsu – Mitsubishi, Sumitomo, DKB, Mitsui, Fuso and Sanwa
  - Mitsubishi & Mitsui
    - Core banks place nearly 200 representatives on the board of other keiretsu members, choose CEOs of about 20% of the member firms

TYPES OF KEIRETSU

Horizontal *keiretsu* (*kinyûkeiretsu)*	
Affiliated `brother and sister` companies spanning different industries	
Vertical keiretsu	
Manufacturing keiretsu (sangyô) Assembled parts move up to the parent through a pyramid of supply companies	**Distribution keiretsu (ryûtsû)** The parent sends finished goods down through a pyramid of distributors to retailers

Mitsubishi Keiretsu

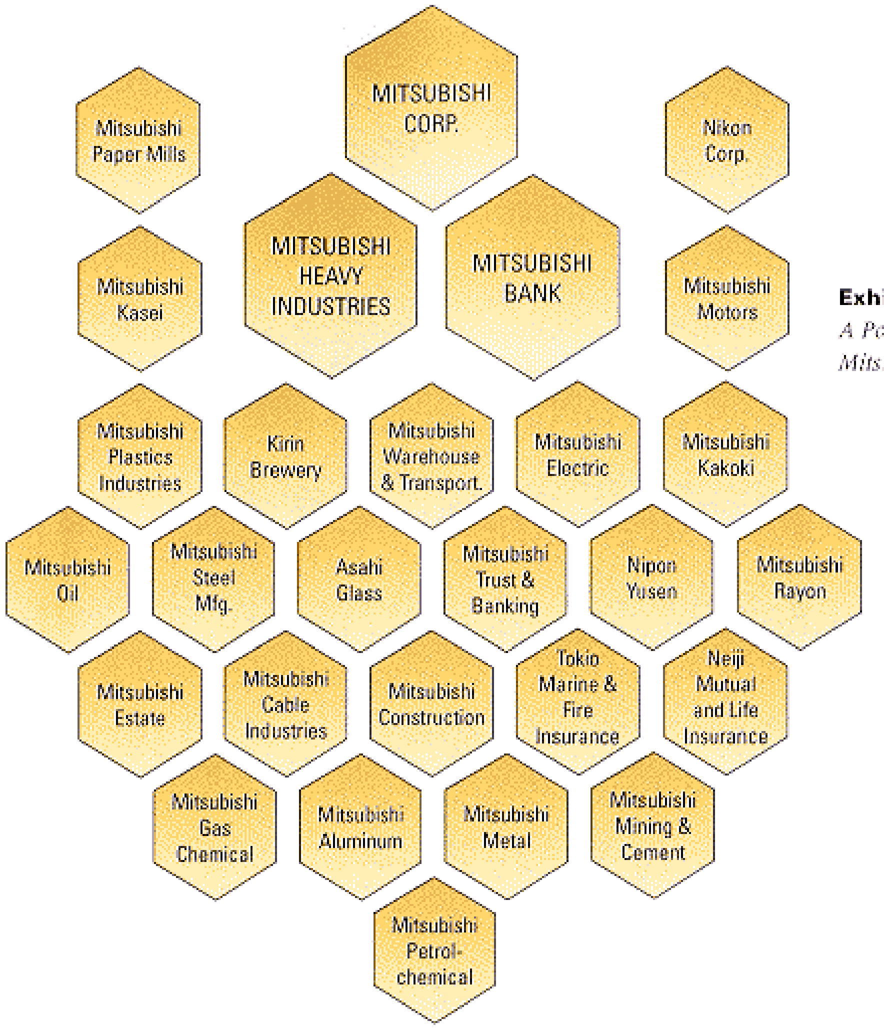


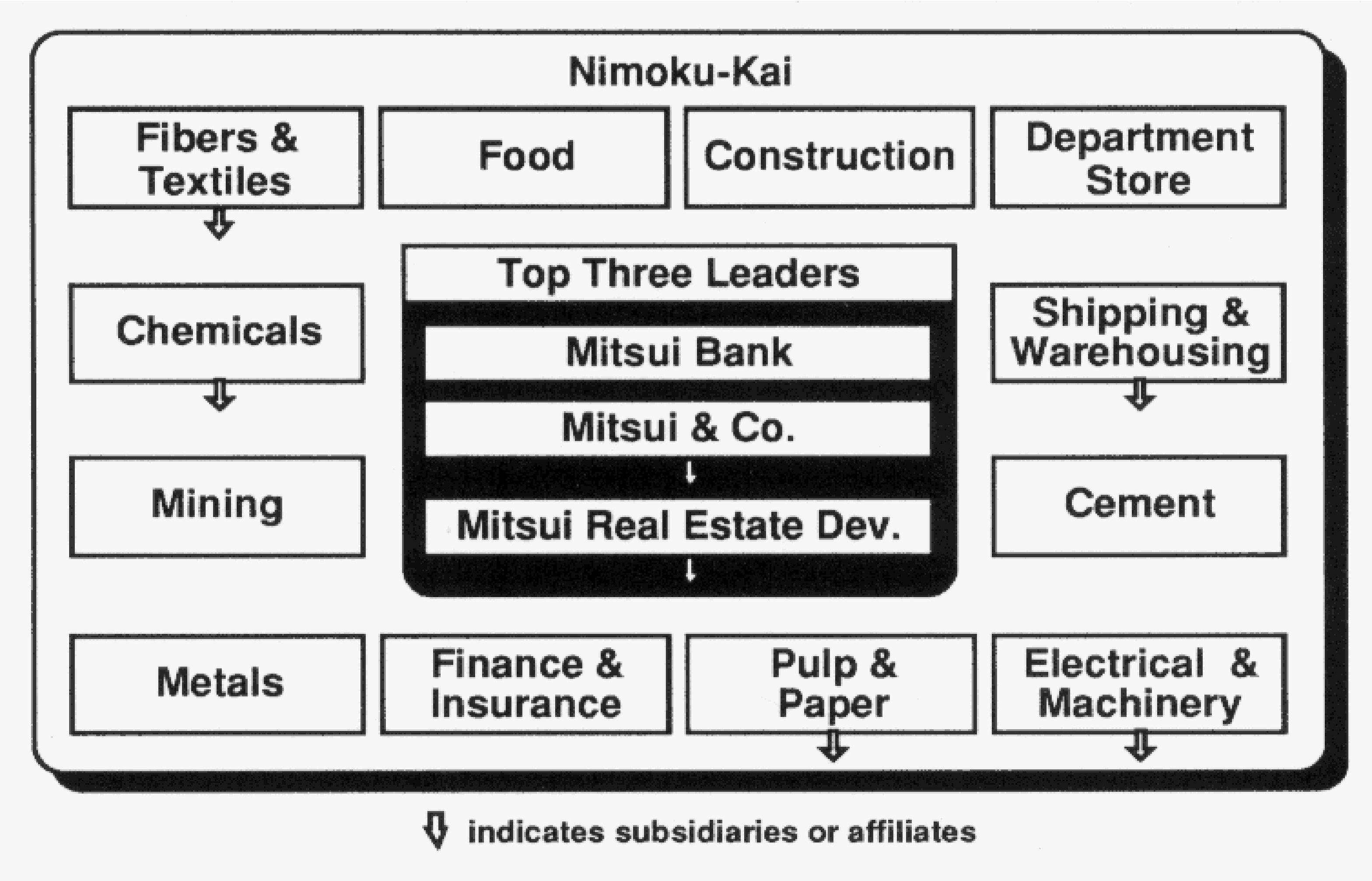
Exhibit 14.2
A Portion of the Mitsubishi Keiretsu

Mitsui keiretsu

TOSHIBA
Leading Innovation >>>

Mitsui Fudosan

MITSUI KEIRETSU



AMERICAN KEIRE

Keiretsu: Content of the Relational Contract

- Where a keiretsu member involved in production experiences a sharp upsurge in demand, it is understood that suppliers in the keiretsu will increase their production without raising supply prices.
- If one keiretsu member requires a product change, it is understood that suppliers will participate in the design discussions
- In a keiretsu, each worker learns each of the others´ businesses, inventory needs, marketing practices, …
- Easy coordination, no short term gains from one member w.r.t. others, lower chance of holdups

VERTICAL INTEGRATION: SUMMARY

- Default: Use Market (BUY)
  - Take advantage of high-powered market incentives!
  - Would your competitor want to buy from you?
    - Do you have any competitive advantage in this business?

- Integrate (MAKE) if:
  - If difficult to write good “performance” contracts
    - YGWYPF: Market provides strong incentives but only for what the market rewards
    - Can you contract on coordination, cooperation?
  - If potential hold-up results in large transactions costs
    - Asset specificity
    - Uncertainty/Complexity => contracting difficult